

Computational Finance

Lecture 9 – American Options

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American Call and Put

- An *American option* is like a European option except that the holder may exercise at any time between the start date and the expiry date.
- A buyer of an American call with maturity T and strike K can exercise the call at any time prior to T . The payoff of exercising the call at time t is

$$(S_t - K)^+.$$

- A buyer of an American put with maturity T and strike K can exercise the put at any time prior to T . The payoff of exercising the put at time t is

$$(K - S_t)^+.$$

- Therefore, a holder of an American option at time $t < T$ needs to decide whether it is worthwhile to exercise the option or continue to hold it.

Equivalence of European and American Call

- In fact, it is never optimal for a holder of an American call to exercise the option before maturity T when there is no dividend payment.
- Exercising the call at time $t < T$ gives the payoff $(S_t - K)^+$. Note that this is only worthwhile if $S_t > K$.
- Alternatively, the option holder may short the underlying asset (which gives the payoff S_t), and purchase the asset by the most favorable of
 - exercising the American call at maturity, and
 - buying at the market price at time T .
- With the alternative strategy the holder has gained an amount S_t at time t and paid out an amount less than or equal to K at time T . This is clearly better than gaining $S_t - K$ at time t .
- Since it is never optimal to exercise an American call option before the expiry date, an American call option must have the same value as a European call option.

Black–Scholes for an American Put (1/3)

- We write $P_{\text{Am}}(S, t)$ to denote the American put option value at asset price S and time t .
- Our first observation is that

$$P_{\text{Am}}(S, t) \geq \max(K - S_t, 0).$$

Note that if $P_{\text{Am}}(S, t) < \max(K - S_t, 0)$, an instantaneous profit can be made by purchasing the option and immediately exercising it.

- Next consider the hedging portfolio $\Pi(S, t) = A(S, t)S + D(S, t)$, where $A = \frac{\partial P_{\text{Am}}(S, t)}{\partial S}$, following the same steps as constructing a hedging portfolio for a European option (when deriving BS PDE).
- In the case of European options, the delta-hedged position is riskless because delta-hedging eliminates all the randomness in the Black–Scholes model, so that we have

$$\Delta(P_{\text{AM}} - \Pi) = r(P_{\text{AM}} - \Pi)\Delta t.$$

- However, it is different for an American put because delta-hedging does not remove the uncertainty associated with early exercise.

Black–Scholes for an American Put (2/3)

- If $\Delta(P_{\text{AM}} - \Pi) > r(P_{\text{AM}} - \Pi) \Delta t$, then $\Delta(P_{\text{AM}} - \Pi)$ does better than cash in the bank.
 - An arbitrageur borrows cash at the cost of r and buys $(P_{\text{AM}} - \Pi)$ at time t , i.e., buying the American put at P_{AM} in the market and selling Π .
 - At time $t + \Delta t$ the arbitrageur sells $(P_{\text{AM}} - \Pi)$ and pays back the cash borrowed at time t . The net profit is $\Delta(P_{\text{AM}} - \Pi) - r(P_{\text{AM}} - \Pi) \Delta t$.
 - The arbitrageur buys the put and therefore controls the exercise facility.
 - In the end $\Delta(P_{\text{AM}} - \Pi) > r(P_{\text{AM}} - \Pi) \Delta t$ will be arbitrated away.
- If $\Delta(P_{\text{AM}} - \Pi) < r(P_{\text{AM}} - \Pi) \Delta t$, then $\Delta(P_{\text{AM}} - \Pi)$ does worse than cash in the bank.
 - Can the arbitrageur profit by selling $(P_{\text{AM}} - \Pi)$ and buying the cash deposit?
 - Suppose at time t the arbitrageur short-sells $(P_{\text{AM}} - \Pi)$ and deposits the cash in the bank account. The arbitrageur enters into a short position of the American put.
 - At time $t + \Delta t$, if the American put is not exercised, the arbitrageur could close the short position and earn $r(P_{\text{AM}} - \Pi) \Delta t - \Delta(P_{\text{AM}} - \Pi)$. However, because the American option could be exercised during $[t, t + \Delta t]$, there is no longer a guarantee of a risk-free arbitrage opportunity.

Black–Scholes for an American Put (3/3)

- These two arguments lead to $\Delta(P_{\text{AM}} - \Pi) \leq r(P_{\text{AM}} - \Pi) \Delta t$. That is,

$$\left[\frac{\partial P_{\text{AM}}}{\partial t} - rD + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 P_{\text{AM}}}{\partial S^2} \right] \Delta t \leq r \left[P_{\text{AM}} - \frac{\partial P_{\text{AM}}}{\partial S} S - D \right] \Delta t,$$

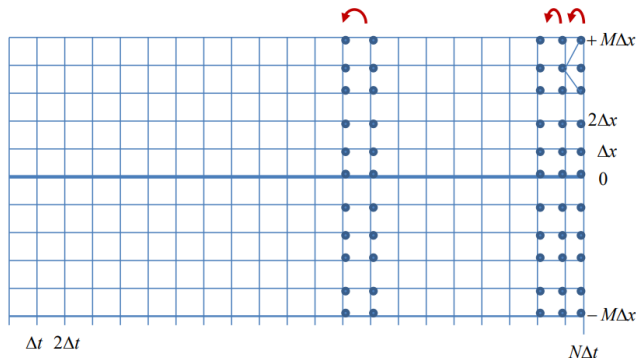
$$\frac{\partial P_{\text{AM}}}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 P_{\text{AM}}}{\partial S^2} + \frac{\partial P_{\text{AM}}}{\partial S} S - rP_{\text{AM}} \leq 0.$$

This is a partial differential inequality.

- In summary:
 - At any point (S, t) , it will be optimal to either exercise or hold on to the option. Hence one of the two below is at equality:
 - $P_{\text{Am}}(S, t) \geq \max(K - S_t, 0),$
 - $\frac{\partial P_{\text{AM}}}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 P_{\text{AM}}}{\partial S^2} + \frac{\partial P_{\text{AM}}}{\partial S} S - rP_{\text{AM}} \leq 0.$
 - The above is called a *linear complementarity problem*.
- The final-time condition is $P_{\text{Am}}(S, T) = \max(K - S_T, 0)$.
- The boundary conditions are $P_{\text{Am}}(0, t) = K$ for $0 \leq t \leq T$, and $\lim_{S \rightarrow \infty} P_{\text{Am}}(S, t) = 0$ for $0 \leq t \leq T$.

Finite Difference Method to Price an American Put (1/2)

- Set up time lattices, $t_n = n\Delta t$, for $n = 1, 2, \dots, N$ with $T = N\Delta t$.
- On each time lattice, discretize the space (e.g. $X = \ln(S_t/K)$) dimension, i.e. $x_m = m\Delta x$ for $m = -M, \dots, M$.



Finite Difference Method to Price an American Put (2/2)

- Initialize the final condition and boundary conditions:

$$V_N^m = \max(K - S_N^m, 0) \quad \text{for } m = -M, \dots, M,$$

$$V_n^M = 0 \quad \text{for } n = 1, 2, \dots, N, \quad V_n^{-M} = K \quad \text{for } n = 1, 2, \dots, N.$$

- For $0 \leq n \leq N - 1$, sequentially calculate the option value at each preceding node (here we enforce the requirement that, at each node, the value of the option is greater than the payoff):

$$V_n^m = \max(K - S_n^m, g(V_{n+1}^{m+1}, V_{n+1}^m, V_{n+1}^{m-1})) \quad \text{for } m = -M + 1, \dots, M - 1,$$

where $g(a, b, c)$ is the stencil function corresponding to the BTCS discretization of the BS formula.

- V_0 (the option price at time 0) is obtained in the final step of the backward loop at x_0 .

Optimal Exercise Boundary

- If S is large, it cannot be worthwhile to exercise an American put $\Rightarrow$ it is optimal to hold on to the option until expiry.
- On the other hand, in the limit $S \rightarrow 0$, the payoff from exercising approaches the maximum possible value that we can attain; it is optimal to exercise.
- There exists a well-defined *optimal exercise boundary* $S^*(t)$, such that
 - for $S_t < S^*(t)$ it is optimal to exercise, so $P_{\text{Am}}(S, t) = K - S_t$, and
 - for $S_t > S^*(t)$ it is optimal to hold, so $P_{\text{Am}}(S, t) > K - S_t$.

Mathematical Formulation of the Pricing Problem

- Let t_0 denote the initial time and $\zeta = \{t_1, \dots, t_N\}$ be the collection of all exercise dates with $\Delta t \equiv t_n - t_{n-1}$, $t_0 < t_1 < \dots < t_N = T$.
- The pricing formula for a Bermudan option with N exercise dates then reads, for $n = N, N-1, \dots, 2$:

$$\begin{aligned}c(x, t_{n-1}) &= \exp(-r\Delta t) \mathbb{E} \left[v(X(t_n), t_n) \mid X(t_{n-1}) = x \right], \\v(x, t_{n-1}) &= \max(\Lambda(x, t_{n-1}), c(x, t_{n-1})),\end{aligned}$$

where

$$v(x, t_0) = \exp(-r\Delta t) \mathbb{E} \left[v(X(t_1), t_1) \mid X(t_0) = x \right].$$

- Here x and y are state variables, defined as

$$X(t_{n-1}) \equiv \ln \left(\frac{S(t_{n-1})}{K} \right).$$

- $v(x, t)$, $c(x, t)$ and $\Lambda(x, t)$ are the option value, the continuation value and the payoff at time t , respectively.
- $v(x, T) = \max(\alpha K(e^x - 1), 0)$, with $\alpha = 1$ for a call and -1 for a put.

MC Method for Pricing American Options (1/10)

- Straightforward MC is computationally infeasible: at each node of one path we need to decide whether to exercise the option or not, whereas the continuation value is not known without either embedding another MC simulation at each node of each time lattice or a prior exercise strategy, since

$$\begin{aligned}c(x, t_{n-1}) &= \exp(-r\Delta t) \mathbb{E} \left[v(X(t_n), t_n) \mid X(t_{n-1}) = x \right], \\v(x, t_{n-1}) &= \max(\Lambda(x, t_{n-1}), c(x, t_{n-1})).\end{aligned}$$

- Industry preference: *Least Squares Monte Carlo* (LSMC), [Longstaff & Schwartz](#).
 - The continuation value, or the conditional expectation, is calculated via least-squares regression.

MC Method for Pricing American Options (2/10)

- **Least squares MC – how does it work?**
- Consider an American put option on a share of non-dividend-paying stock. The put option is exercisable at strike price 1.10 at times 1, 2, and 3, where time 3 is the final expiration date. The riskless rate is 6%.
- For simplicity, we illustrate the algorithm using only eight sample paths for the price of the stock. These sample paths are generated under the risk-neutral measure and are shown in the following matrix.

Path	Stock price paths			
	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1.00	1.09	1.08	1.34
2	1.00	1.16	1.26	1.54
3	1.00	1.22	1.07	1.03
4	1.00	.93	.97	.92
5	1.00	1.11	1.56	1.52
6	1.00	.76	.77	.90
7	1.00	.92	.84	1.01
8	1.00	.88	1.22	1.34

MC Method for Pricing American Options (3/10)

- Conditional on not exercising the option before the final expiration date at time 3, the cash flows realized by the option holder from following the optimal strategy at time 3 are given below. These cash flows are identical to the cash flows that would be received if the option were European rather than American.

Stock price paths				
Path	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1.00	1.09	1.08	1.34
2	1.00	1.16	1.26	1.54
3	1.00	1.22	1.07	1.03
4	1.00	.93	.97	.92
5	1.00	1.11	1.56	1.52
6	1.00	.76	.77	.90
7	1.00	.92	.84	1.01
8	1.00	.88	1.22	1.34

Cash-flow matrix at time 3			
Path	$t = 1$	$t = 2$	$t = 3$
1	—	—	.00
2	—	—	.00
3	—	—	.07
4	—	—	.18
5	—	—	.00
6	—	—	.20
7	—	—	.09
8	—	—	.00

MC Method for Pricing American Options (4/10)

- If the put is in the money at time 2, the option holder must then decide whether to exercise the option immediately or continue the option's life until the final expiration date at time 3.
- From the stock-price matrix, there are only five paths for which the option is in the money at time 2. Let X denote the stock prices at time 2 for these five paths and Y denote the corresponding discounted cash flows received at time 3 if the put is not exercised at time 2.

Stock price paths				
Path	$t = 0$	$t = 1$	$t = 2$	$t = 3$
1	1.00	1.09	1.08	1.34
2	1.00	1.16	1.26	1.54
3	1.00	1.22	1.07	1.03
4	1.00	.93	.97	.92
5	1.00	1.11	1.56	1.52
6	1.00	.76	.77	.90
7	1.00	.92	.84	1.01
8	1.00	.88	1.22	1.34

Regression at time 2		
Path	Y	X
1	$.00 \times .94176$	1.08
2	—	—
3	$.07 \times .94176$	1.07
4	$.18 \times .94176$	.97
5	—	—
6	$.20 \times .94176$	.77
7	$.09 \times .94176$	.84
8	—	—

MC Method for Pricing American Options (5/10)

- To estimate the expected cash flow from continuing the option's life conditional on the stock price at time 2, we regress Y on a constant, X , and X^2 . The resulting conditional expectation function is

$$\mathbb{E}[Y | X] = -1.070 + 2.983 X - 1.813 X^2.$$

- With this conditional expectation function, we now compare the value of immediate exercise at time 2 (left): it is optimal to exercise the option at time 2 for the fourth, sixth, and seventh paths. This leads to the cash-flow at time 2 (right).

Optimal early exercise decision at time 2

Path	Exercise	Continuation
1	.02	.0369
2	—	—
3	.03	.0461
4	.13	.1176
5	—	—
6	.33	.1520
7	.26	.1565
8	—	—

Cash-flow matrix at time 2

Path	$t = 1$	$t = 2$	$t = 3$
1	—	.00	.00
2	—	.00	.00
3	—	.00	.07
4	—	.13	.00
5	—	.00	.00
6	—	.33	.00
7	—	.26	.00
8	—	.00	.00

MC Method for Pricing American Options (6/10)

- Proceeding recursively, we next examine whether the option should be exercised at time 1. From the stock-price matrix, there are again five paths where the option is in the money at time 1. For these paths, we again define Y as the discounted value of subsequent option cash flows.
- Since the option can only be exercised once, future cash flows occur at either time 2 or time 3, but not both. Cash flows received at time 2 are discounted back one period to time 1, and any cash flows received at time 3 are discounted back two periods to time 1.

Regression at time 1		
Path	Y	X
1	$.00 \times .94176$	1.09
2	—	—
3	—	—
4	$.13 \times .94176$	.93
5	—	—
6	$.33 \times .94176$	.76
7	$.26 \times .94176$	.92
8	$.00 \times .94176$	.88

Cash-flow matrix at time 2			
Path	$t = 1$	$t = 2$	$t = 3$
1	—	.00	.00
2	—	.00	.00
3	—	.00	.07
4	—	.13	.00
5	—	.00	.00
6	—	.33	.00
7	—	.26	.00
8	—	.00	.00

MC Method for Pricing American Options (7/10)

- The conditional expectation function at time 1 is estimated by again regressing Y on a constant, X and X^2 . The estimated conditional expectation function is

$$\mathbb{E}[Y | X] = 2.038 - 3.335 X + 1.356 X^2.$$

- Substituting the values of X into this regression gives the estimated conditional expectation function. Comparison shows that exercise at time 1 is optimal for the fourth, sixth, seventh, and eighth paths.

Optimal early exercise decision at time 1

Path	Exercise	Continuation
1	.01	.0139
2	—	—
3	—	—
4	.17	.1092
5	—	—
6	.34	.2866
7	.18	.1175
8	.22	.1533

MC Method for Pricing American Options (8/10)

- Having identified the exercise strategy at times 1, 2, and 3, the exercise rule can now be represented by the following matrix, where the ones denote exercise dates at which the option is exercised.

Stopping rule			
Path	$t = 1$	$t = 2$	$t = 3$
1	0	0	0
2	0	0	0
3	0	0	1
4	1	0	0
5	0	0	0
6	1	0	0
7	1	0	0
8	1	0	0

Option cash flow matrix			
Path	$t = 1$	$t = 2$	$t = 3$
1	.00	.00	.00
2	.00	.00	.00
3	.00	.00	.07
4	.17	.00	.00
5	.00	.00	.00
6	.34	.00	.00
7	.18	.00	.00
8	.22	.00	.00

- Determine the cash flows realized by following this stopping rule: simply exercise the option at the exercise dates where there is a one in the above matrix.
- The option can now be valued by discounting each cash flow in the option cash-flow matrix back to time zero, and averaging over all paths.
 - It results in a value of 0.1144 for the American put. This is roughly twice the value of 0.0564 for the European put obtained by discounting back the cash flows at time 3.

MC Method for Pricing American Options (9/10)

- Now let's formulate this approach mathematically:
 - No-arbitrage valuation theory implies that the value of continuation $F(\omega; t_k)$, or equivalently, the value of the option assuming that it cannot be exercised until after t_k , is given by taking the expectation of the remaining discounted cash flows $C(\omega, s; t_k, T)$ w.r.t. the risk-neutral pricing measure Q , i.e.

$$F(\omega; t_k) = \mathbb{E}^Q \left[\sum_{j=k+1}^N \exp \left(- \int_{t_k}^{t_j} r(\omega, s) ds \right) C(\omega, t_j; t_k, T) \mid \mathcal{F}_{t_k} \right].$$

- The LSM approach uses least squares to approximate the conditional expectation function at $t_{N-1}, t_{N-2}, \dots, t_1$. This is done backwards in time:
 - At time t_{N-1} , assume $F(\omega; t_{N-1})$ can be represented as a linear combination of a countable set of $\mathcal{F}_{t_{N-1}}$ -measurable basis functions:

$$F(\omega; t_{N-1}) = \sum_{j=0}^{\infty} a_j L_j(X).$$

MC Method for Pricing American Options (10/10)

- Now let's formulate this approach mathematically (continued):
 - Once the conditional expectation function at time t_{N-1} is estimated, we can determine whether early exercise at time t_{N-1} is optimal for an in-the-money path ω .
 - Once the exercise decision is identified, the option cash-flow paths $C(\omega, t_j; t_{N-2}, T)$ can then be approximated.
 - The recursion proceeds by rolling back to time t_{N-2} and repeating the procedure until the exercise decisions at each exercise time along each path have been determined.
 - The American option is then valued by starting at time zero, moving forward along each path until the first stopping time occurs, discounting the resulting cash flow from exercise back to time zero, and then taking the average over all paths.

The COS Method for Pricing American Options (1/3)

- The COS method has no problem finding the conditional expectation as it makes use of the distribution information directly.
- Start from the mathematical formulation of the pricing problem:

$$c(x, t_{n-1}) = \exp(-r\Delta t) \mathbb{E}[V(X(t_n)) | X(t_{n-1}) = x] = \exp(-r\Delta t) \int_{-\infty}^{\infty} v(y, t_n) f(y | x) dy,$$
$$v(x, t_{n-1}) = \max(\Lambda(x, t_{n-1}), c(x, t_{n-1})).$$

- Replace $c(x, t_{n-1})$ by the COS approximation, i.e.

$$c(x, t_{n-1}) \approx \hat{c}(x, t_{n-1}) \equiv e^{-r\Delta t} \sum_{k=0}^K \operatorname{Re} \left[\varphi \left(\frac{k\pi}{b-a} \right) e^{ik\pi(x-a)/(b-a)} \right] V_k(t_n),$$

where

$$V_k(t_n) = \begin{cases} C_k(a, x_n^*, t_n) + G_k(x_n^*, b) & \text{for a call,} \\ G_k(a, x_n^*) + C_k(x_n^*, b, t_n) & \text{for a put,} \end{cases}$$
$$G_k(x_1, x_2) \equiv \frac{2}{b-a} \int_{x_1}^{x_2} \Lambda(x, t_n) \cos\left(\frac{k\pi(x-a)}{b-a}\right) dx,$$
$$C_k(x_1, x_2, t_n) \equiv \frac{2}{b-a} \int_{x_1}^{x_2} c(x, t_n) \cos\left(\frac{k\pi(x-a)}{b-a}\right) dx.$$

- Further, replacing $c(x, t_n)$ by $\hat{c}(x, t_n)$ in the definition above of $V_k(t_n)$, we obtain $\hat{V}_k(t_n)$.

The COS Method for Pricing American Options (2/3)

- **Algorithm:**

- *Initialization:* For $k = 0, 1, \dots, K - 1$, $V_k(t_N) = G_k(0, b)$ for call options and $G_k(a, 0)$ for put options.
- *Main loop* to recover $\hat{V}_k(t_n)$, for $n = N - 1, \dots, 1$:
 - Determine the early-exercise point x_n^* by Newton's method;
 - Compute $\hat{V}_k(t_n)$ with the help of the FFT algorithm.
- *Final step:* reconstruct $\hat{v}(x, t_0)$.

- **Computational complexity:**

- $NK \log_2(K)$.

The COS Method for Pricing American Options (3/3)

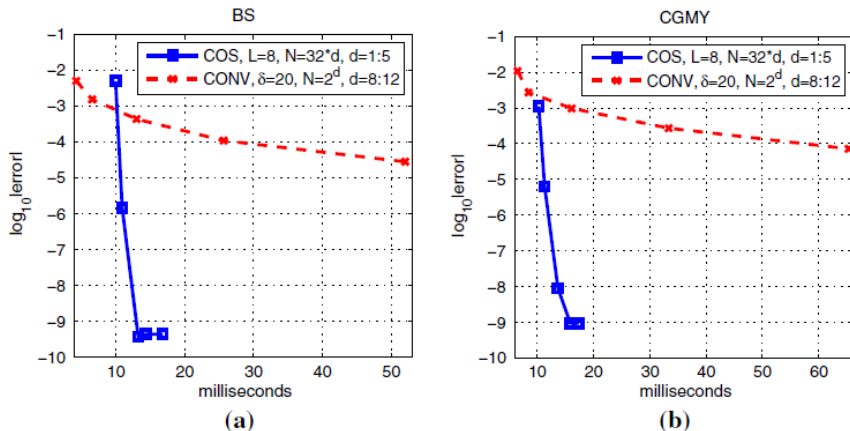


Fig. 2 Error versus CPU time for pricing Bermudan put options under **a** BS (Test no. 1) and **b** CGMY ($Y = 1.5$, Test no. 2) model, comparing the COS and the CONV method

Richardson Extrapolation: From Bermudan to American

- All the numerical methods effectively price a Bermudan option, as the monitoring dates are from a discrete grid.
- As the number of monitoring dates increases, the price converges to that of an otherwise identical American option.
- Note that there is a faster way to get prices of American options: applying a *Richardson extrapolation* on the prices of a few Bermudan options.
 - Let $v(M)$ denote the value of a Bermudan option with M early-exercise dates. A 4-point Richardson extrapolation scheme reads

$$v_{\text{AM}}(d) = \frac{1}{21} (64 v(2^{d+3}) - 56 v(2^{d+2}) + 14 v(2^{d+1}) - v(2^d)) .$$

Appendix: Binomial Method for Pricing American Options

- It turns out that a straightforward adaptation of the binomial method can be used to value an American put. While working backward through the tree, the option value at the node S_m^n is the maximum of the exercise value and the hold value.
- The algorithm is as follows:
 - Initialize the option value at expiry $T = N\Delta t$. The option value at the m -th node of expiry is simply given by the payoff function, i.e.

$$V_m^N = \max(K - S_m^N, 0), \quad 0 \leq m \leq M.$$

- For $0 \leq n \leq N - 1$, sequentially calculate the option value at each preceding node, i.e.

$$V_m^n = \max(K - S_m^n, e^{-r\Delta t} [p V_{m+1}^{n+1} + (1 - p) V_m^{n+1}]), \quad 0 \leq m \leq n.$$

- V_0 (the option price at time 0) is obtained in the final step of the backward loop.